Group Project Checkpoint: Proposal of Final Project Design

Introduction

Each team will submit a written proposal that documents the planned design of your Final Project. The proposal should demonstrate that you have identified a compelling business problem, defined clear objectives, outlined your data and modeling strategies, and anticipated system-level considerations that will be refined later in the course. This is a project plan, not a progress report.

📄 Format Guidelines

- Document: Word or PDF
- Length: ~6 pages (excluding cover, index, references)
- Font: Calibri, 11 pt
- Line spacing: 1.15
- Numbered sections + table of contents

Required Sections:

1. Executive Summary (0.5–1 page)

In this section, provide a clear overview of your planned project:
- What problem are you solving?
- Why is it important?
- Who are the users or stakeholders?
- What outcome or impact do you expect?

Tip: Remember that ML projects often fail when they are not tied to real-world value. Make that connection explicit here.

2. Business Case & Objectives (1 page)

Describe the planned value of your project:
- Define business objectives (value creation, ROI, cost reduction, user impact).
- Define ML objectives (performance metric, latency, interpretability).
- Show how your ML objectives support the business case.
- State your success criteria (e.g., target metric or business KPI).

3. Scope, Constraints & Risks (0.5–1 page)

Outline the boundaries of your project:
- In-scope / out-of-scope tasks.
- Constraints (time, compute, data access, labeling resources).
- Risks (data imbalance, bias, unrealistic expectations).
- Trade-offs you anticipate (false positives vs false negatives, interpretability vs accuracy).

4. Data Plan (1–1.5 pages)

Explain your data strategy:
- Data sources (planned datasets: public, synthetic, internal).
- Data quality (completeness, biases, representativeness).
- Labeling (planned process, challenges, disagreements).
- Sampling/imbalance considerations.
- How you will avoid pitfalls such as data imbalance, label multiplicity, or leakage.

5. Modeling Approach (1.5–2 pages)

Outline your modeling plan:
- Baselines (rules, heuristics, simple models). Define your usefulness threshold.
- Candidate models: which algorithms you will test first and why.
- Feature engineering ideas (planned transformations, augmentations, domain-specific features).
- Evaluation metrics: which metrics matter and why.
- Interpretability: whether your system needs to be explainable at this stage or later.

6. System Preview & Project Plan (1 page)

You are not expected to provide technical details about deployment or monitoring yet. Instead, sketch your initial vision:
- A simple block diagram: data → preprocessing → model → output.
- Where predictions might be consumed (report, app, API, dashboard).
- Timeline & milestones (data collection, baseline, first model, evaluation, final demo).
- Team roles (who focuses on data, features, modeling, documentation). Roles can overlap, but all areas should be covered.

7. References

List datasets, papers, repositories, or prior work you consulted. Format all references in APA 7th edition style.

Appendix: Grading Rubric

This rubric will be used to grade the Group Project Checkpoint. Each section has a maximum number of points, summing to 100. Performance descriptions below clarify what is expected:

Section	Max Points	Excellent	Adequate	Weak
Executive Summary	10	Problem clearly defined, importance justified, stakeholders identified, outcome realistic and ambitious.	Problem somewhat clear, limited justification, vague stakeholders or outcome.	Problem unclear, little/no justification, missing stakeholders or outcomes.
Business Case & Objectives	25	Compelling business value, ML objectives tightly aligned with goals, success criteria measurable and meaningful.	Business value partially defined, ML objectives loosely connected, metrics incomplete.	Business case vague, no clear objectives or metrics.
Scope, Constraints & Risks	15	Scope realistic and well delimited, feasible constraints, key risks identified with mitigations.	Scope partially defined, some constraints vague, risks noted but not detailed.	Scope missing or unrealistic, no constraints or risks.
Data Plan	15	Relevant datasets identified, quality & bias discussed, labeling/imbalance strategy clear.	Some datasets identified, limited discussion of quality or labeling.	No clear datasets, ignores quality, labeling, or imbalance.
Modeling Approach	15	Strong baselines, justified candidate models, relevant evaluation metrics, thoughtful features.	Some baselines/models, limited justification, vague metrics/features.	No baselines, unjustified models, missing metrics/features.
System Preview & Project Plan	10	Clear block diagram, realistic timeline, well-defined team roles and milestones.	Diagram or timeline incomplete, vague roles.	Missing diagram/timeline, roles undefined.
Presentation & Writing	10	Well-structured document around 6 pages (exc. cover,	Acceptable structure, minor formatting/writing	Poor structure, wrong length,

		index, references), polished writing, professional formatting, proper references.	issues, some references.	unclear writing, missing references.